

is considered to be Martin Cooper, a Motorola researcher. In 1973, he filed a patent for a “radio telephone system”. This kicked off the era of the mobile phone. By 1979, NTT of Japan had a fully operational city-wide cellular network.

The USA’s first approved mobile phone, the Motorola Dynatac, appeared in 1983. By 1984 Bell labs had developed the technology of the cellular phone, the ancestor to the modern cell phone. This technology relied on multiple base stations or cells (towers) each covering a small area. The switching technology required to ensure that a telephone conversation can continue as a cell phone moves from one cell to another was also developed then. By the late 1980s, mobile phones based on this technology, known as 1G (first generation) mobile phones were quite popular in the US, Japan and the Scandinavian countries. The first generation differed from modern mobile phones in that 1G phones were essentially analogue. The voice signals were not encoded, merely modulated to a higher frequency.

The first mobile phone essentially used the same technology as is popular today. The 2G digital standard, was launched in Finland in 1991. This utilised digital signals and was based on the now popular GSM technology. SMS was introduced in 1993 in Finland. Ever since then, mobile phones have not looked back. 2G gave way to 2.5G and now 3G.

Today, the mobile phone technologies used in India can be divided into two - GSM and CDMA.



The first practical cellphone, the Motorola DynaTac 3000X

4.2 GSM

The leading mobile phone standard in use in the world today, GSM is the technology used by providers such as Airtel, Vodafone, Idea and Aircel. The advantages of this system come from the fact that most providers worldwide use this, resulting in easy interoperability of service providers. This allows services